



C u s t o m e r B e h a v i o r D a s h b o a r d

A data-driven exploration of purchasing patterns, customer segments, and actionable retail insights

PORTFOLIO PROJECT

DATA ANALYTICS



Project Overview

🎯 Goal

Uncover hidden patterns in customer purchasing behavior to drive smarter business decisions

📊 Approach

End-to-end pipeline: data ingestion → cleaning → EDA → SQL analysis → interactive dashboard

📈 Outcome

Interactive dashboard with segmented insights and strategic recommendations for stakeholders

P r o b l e m



The Challenge

Retail businesses lack visibility into **who buys what, when, and why** — leading to missed revenue and poor targeting.

O b j e c t i v e s

- Identify top customer segments and spending patterns
- Analyze product performance and category trends
- Measure repeat purchase behavior and loyalty signals
- Deliver actionable recommendations for marketing teams

D a t a s e t

Customers	
	demographics
	age
	gender
	location

Transactions	
	order ID
	date
	payment method

Products	
	category
	price
	brand

Order Items	
	quantity
	unit price
	discount

W h a t '

50,000+ Transactions

Spanning 12 months of retail activity

4 Relational Tables

Customers, Orders, Products, Order Items

8

Age, Gender, Location, Category, Payment, Season, Discount, Frequency

T o o l s



P y t h o n

Pandas, NumPy, Matplotlib — data manipulation & visualization



S Q L

PostgreSQL — complex queries, aggregations, window functions



P o w e r

Interactive dashboards, DAX measures, drill-through filters

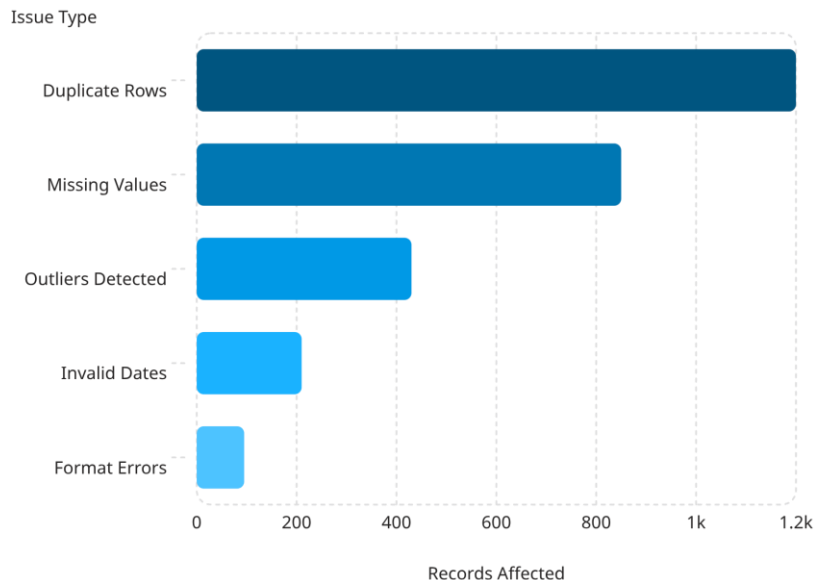


E x c e l

Initial data profiling, pivot tables, quick exploratory checks



Data



Cleaning Pipeline

01

Remove Duplicates

1,200 redundant transaction records eliminated

02

Handle

Imputed or removed 850 missing entries

03

Outlier Treatment

IQR method applied to price & quantity fields

ÇÇ

Standardize

Unified date, currency, and categorical fields

S Q L

T o p

R F M

S e g m e n t a t i o n

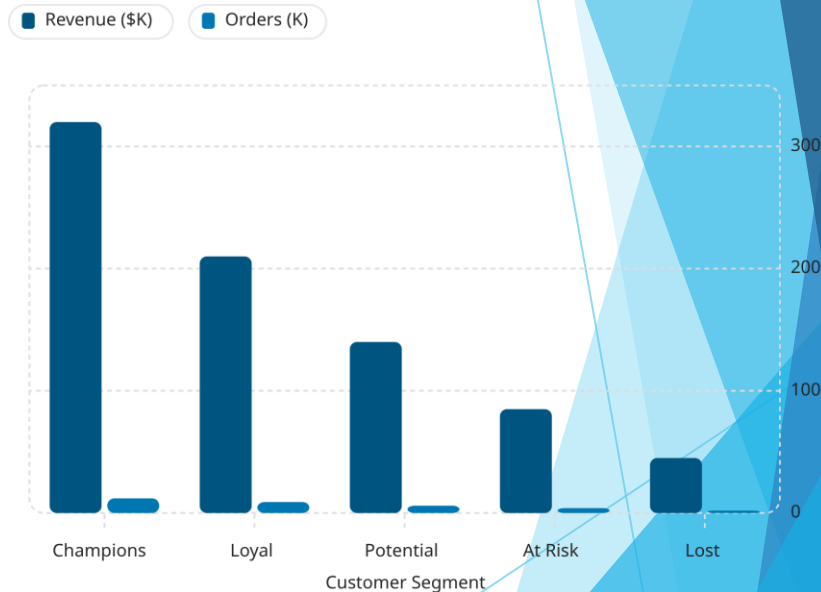
Recency, Frequency,
Monetary scoring per
customer

C o h o r t

Retention rates by
acquisition month

C a t e g o r y

Revenue breakdown with MoM growth



D a s h b o a r d



K P I

Total Revenue, AOV, Repeat Rate, New vs. Returning

Geo Map

Sales density by region with drill-down capability

Time Trends

Monthly revenue, seasonality, and peak shopping days

S e g m e n t s

RFM matrix with interactive customer group filtering

Key



2
customers
6

Prioritize retention for
Champions & Loyal
segments



Electronics
Fashion
sales

Bundle promotions
across top categories



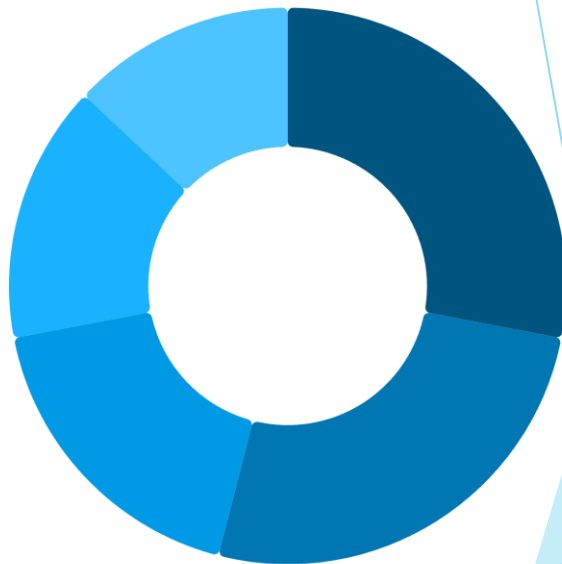
Peak **-9**
P M

Time-targeted push
notifications and flash
sales



38% one-time
buyers at risk

Launch re-engagement
email campaigns with
incentives



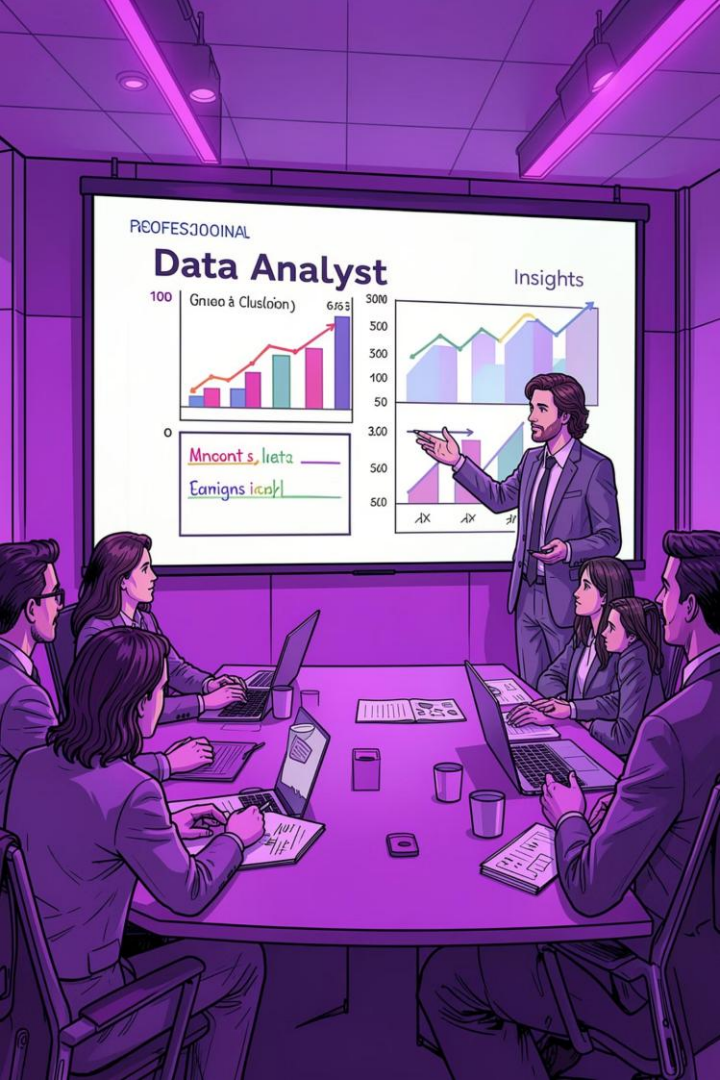
Electronics · 28%

Fashion · 26%

Home & Living · 18%

Beauty · 15%

Sports · 13%



C o n c l u s i o n

This project demonstrates end-to-end analytical capability — from raw data to strategic insight — using industry-standard tools and methodologies.

☒ C l e a n ,

☒ Rigorous SQL
Analysis

☒ A c t i o n a b l e

Thank you for your time.